Research Designs & Participant Assignment
Part 1 — Between-Groups Design: Assignment to Conditions

Imagine that you were a research assistant in the laboratory that was planning to carry out the revised design to further examine "Advantages of Video-Based Software Instruction". The various sign-up sheets have been transferred to a single page that lists the order in which the volunteers will come (one at a time) to the laboratory. Let's examine the ways that participants might be assigned in this between groups design

Assignment to conditions #1

One approach is to flip a coin for each participant – using say … heads = instruction manual & "tails" = instructional video. Apply this approach below – putting "manual" or "video" beside each participant's name under Assignment #1

For statistical reasons it is convenient that we have "=n". That is, the same number of participants in each condition of the between groups research design. However this doesn't always happen when this approach is used.

How many folks in the lab got =n when they used this approach?

What might improve the chances of getting =n when using this approach?

Assignment to conditions #2

The more commonly used approach is the one described in class – the first person is assigned by coin-flip and the second is automatically assigned to the other condition; the third person by coin-flip and the fourth to the opposite, etc. This is generally known as a "randomized block" procedure – a "block" is one participant in each treatment condition. Apply this approach below – putting "manual" or "video" beside each participant's name under Assignment #2

Assignment to conditions #3

One unpleasant reality of data collection is "no shows" – people who "withdraw their voluntary participation before or during completing of the data collection interval". You know – the ones that blow you off!!! The "*" indicate folks who are no-shows. How would you modify the random assignment from #2 to "adjust" for these thoughtless, ungrateful, manipulative, %$^&^$#@@ individuals?

Subject	Assignment #1	Assignment #2	Assignment #3
Smith*			
Jones			
Cladestic			
Worthy*			
Nerop*			

Subject	Assignment #1	Assignment #2	Assignment #3
Blick			
Rethuk			
Iffski			
Testa			
Johnston*			
Harrison			
Beasley			

Part 2 — Initial Equivalence in a Between-Groups Design

Here are the eight folks who have signed up to be in your experiment. The study uses a between groups design with a Tx and a C condition. Randomly assign folks to the two conditions, using a randomized blocks procedure.

Mark the condition to which you randomized each person in the table below.

Let's evaluate the initial equivalence wrought by randomized assignment… Here are some data for each of our participants.

Subject #	Employed	Motivation	Prior Task Experience	IV conditions assigned (C or Tx)
S1	N	10	0	
S2	N	20	1	
S3	Y	20	0	
S4	Y	20	2	
S5	N	10	0	
S6	Y	10	1	
S7	N	10	0	
S8	Y	20	2	

Below, summarize the data for each variable for each of the IV conditions.

	Tx condition	C condition
Employment	# emp ________ # non-emp ________	# emp ________ # non-emp ________
Motivation	average ______________	average ______________
Prior Task Experience	average ______________	average ______________

Check with the other folks in your lab…

How many got an even employment-split in both conditions?

How many got the same average motivation in both conditions?

How many got the same average prior task experience in both conditions?

How many got initial equivalence for all three variables????

So, what do you think … How "guaranteed" is initial equivalence based on proper random assignment ?

What would probably have improved the initial equivalence produced by this (or any other) random assignment?

Part 3 — Within-Groups Design: Assignment to Condition Order

This time, imagine that you were a research assistant in a laboratory that was planning to carry out the revised design to further examine "Advantages of Video-Based Software Instruction" using a within-subjects design. The various sign-up sheets have been transferred to a single page that lists the order in which the volunteers will come (one at a time) to the laboratory. Let's examine the ways that participants might be assigned in this within-groups design

Assignment to condition order #1

One approach is to flip a coin for each participant – using say … heads = complete the instruction manual condition first and then redo the assignment using the video instruction & "tails" = complete the instructional video condition first and then redo the assignment using the instruction manual. Apply this approach below – putting either "M-V" for "manual first then video" or "V-M" beside each participant's name under Assignment #1

To maintain initial equivalence, it is essential that we have "=n" – the same number of participants complete the two conditions in each order. However this doesn't always happen when this approach is used.

How many folks in the lab got =n for the two orders when they used this approach?

What might improve the chances of getting =n for the two orders when using this approach?

Assignment to condition order #2

The more commonly used approach is the one described in class – the first person is assigned to a condition order by coin-flip and the second is automatically assigned to the opposite condition order; the third person by coin-flip and the fourth to the opposite, etc. This is also generally known as a "randomized block" procedure – yes, the same term is used for BG and WG assignment procedures, confusing, huh? Apply this approach below – putting either "M-V" or "V-M" beside each participant's name under Assignment #2

Assignment to condition order #3

One unpleasant reality of data collection is "no shows" – people who "withdraw their voluntary participation before or during completing of the data collection interval". You know – the ones that blow you off!!! The "*" indicate folks who are no-shows. How would you modify the random assignment from #2 to "adjust" for these thoughtless, ungrateful, manipulative, %$^&^$#@@ individuals?

Subject	**Assignment #1**	**Assignment #2**	**Assignment #3**
Smith			
Jones*			
Cladestic*			
Worthy*			
Nerop*			
Blick			
Rethuk*			
Iffski			
Testa			
Johnston*			
Harrison			
Beasley			

By the way – how good of an idea is it to run this study as a within-groups design? Why or why not???

Part 4 — Selecting Between-Groups and Within-Groups Designs

Having brought up the subject of selecting BG and WG designs, let's explore it further.

Make up three causal research questions.

Keep it simple – just 2-conditions to be compared.

One of these should be …

- One question that could "reasonably" be answered using either a BG or a WG design
- One question that could only "reasonably" be answered using a BG design
- One question that could only "reasonably" be answered using a WG design

After you've chosen your questions, we'll have everybody ask their questions and consider the best type of design to use for each.

The goal is to develop a set of "rules" for deciding whether a study should be conducted using a BG or a WG design (& when it doesn't matter).

Your three questions…

What "rules" did you identify?

Assignment grade out of 10 points ________________